

How to treat autism in children?

Evidence searched on 2026-10-09 (PubMed, ClinicalTrials.gov, FDA drug labels, web)

Clinical summary

Bottom line The retrieved evidence supports **intervention-based approaches** for specific domains in autism in children, including **motor skills (exercise, dance, occupational therapy)** and **feeding difficulties (structured, individualized treatment algorithms)**.

Most important recent finding Exercise interventions significantly improved overall, gross, and fine motor skills in children with ASD, with dose-response relationships identified for overall and gross motor skills [1].

Immediate context Evidence is **narrow and intervention-specific**: motor skills and feeding difficulties are supported by structured programs, but no direct evidence exists for broader autism symptom management (e.g., social communication, behavior). FDA-approved drugs (e.g., aripiprazole, risperidone) exist for irritability but are not referenced in the retrieved studies.

Top treatment drugs (established first, then the most notable latest drug)

Drug	Indication named in the FDA label	Source
Aripiprazole	"Aripiprazole is indicated for the treatment of: Irritability Associated with Autistic Disorder."	DailyMed
Risperidone	"Risperidone tablets are indicated for the treatment of irritability associated with autistic disorder, including symptoms of aggression towards others, deliberate self-injuriousness, temper tantrums, and quickly changing moods. Efficacy was established in 3 short-term trials in children and adolescents (ages 5 to 17 years)."	DailyMed
NTI164 (latest, investigational)	Being tested in a registered Phase 3 trial, Not yet recruiting; not an approved treatment	NCT07257939

Latest findings

- **2026-09-22:** Case study, a modified ketogenic diet combined with rehabilitation training showed short-term reductions in ABC and CARS scores in young children with ASD compared to rehabilitation alone [2].
- **2026-09-18:** Meta-analysis, exercise significantly improved overall, gross, and fine motor skills in children with ASD, with dose-response relationships identified for overall and gross motor skills [1].
- **2026-09-11:** Randomized controlled trial, a modified ketogenic diet combined with rehabilitation training demonstrated short-term feasibility and greater reductions in ASD symptom scores than rehabilitation alone [2].
- **2026-09-08:** Clinical trial protocol, a community-based dance program for children with autism is under investigation (ANZCTR12625000600448).

Current evidence

Approach	Evidence level	What the evidence shows	Reference
Exercise interventions	Evidence-supported but limited	Exercise significantly improved overall, gross, and fine motor skills in children with ASD, with dose-response effects.	[1]
Modified ketogenic diet + rehab	Evidence-supported but limited	Short-term reductions in ASD symptom scores (ABC, CARS) compared to rehabilitation alone, with acceptable feasibility.	[2]
Naturalistic Developmental Behavioral Interventions (NDBIs)	Emerging	Caregiver-assisted group-based NDBIs show promise as accessible early interventions, improving developmental and social outcomes in preschoolers.	[3]
Psychological interventions for caregivers	Indirect evidence (different or related population)	Different or related population (see the cited study); not condition-specific	[4]
Feeding difficulties treatment (FEEDSETUP)	Indirect evidence (different or related population)	Different or related population (see the cited study); not condition-specific	[5]
Extended Reality (XR) interventions	Emerging	XR (e.g., VR, AR) shows growing research interest for supporting children with autism, with hybrid systems and training/educational applications leading.	[6]
Deep brain stimulation (DBS)	Experimental or preliminary	Dual-target DBS in a 12-month case study showed sustained reductions in core ASD symptoms (ATEC, RBS-R, AuBC, CARS-2) without adverse events.	[7]

Treatment Drug Landscape

Drugs used for treatment of this condition

Approved: an FDA label names this condition. Guideline-recommended: a guideline organisation's page recommends the drug. Doses appear only where an FDA label or source gives them for the use shown; each dose entry is a short summary of the label's dosing section for this condition, with every number checked against the label text (full text in the DailyMed link); individual dosing is a prescriber's decision.

Drug	Indication	Dose / route (if sourced)	Approval status / date	Key evidence
Aripiprazole	"Aripiprazole is indicated for the treatment of: • Schizophrenia • Irritability Associated with Autistic Disorder • Treatment of Tourette's Disorder" [associated symptom or comorbidity, not the condition itself]	Route: Oral. Pediatric Patients (6 to 17 years) The recommended dosage range for the treatment of pediatric patients with irritability associated with autistic disorder is 5 to 15 mg/day. Dosing should be initiated at 2 mg/day. The dose should be increased to 5 mg/day, with subsequent increases to 10 or 15 mg/day if needed. Dose adjustments of up to 5 mg/day should occur gradually, at intervals of no less than one week.	FDA-approved for the use shown; label effective 2026-03-05; original approval date not retrieved	FDA label. DailyMed label

Drug	Indication	Dose / route (if sourced)	Approval status / date	Key evidence
Risperidone	"Risperidone tablets are indicated for the treatment of irritability associated with autistic disorder, including symptoms of aggression towards others, deliberate self-injuriousness, temper tantrums, and quickly changing moods. Efficacy was established in 3 short-term trials in children and adolescents (ages 5 to 17 years)." [associated symptom or comorbidity, not the condition itself]	Route: Oral. Start with 0.25 mg per day for children weighing less than 20 kg, and 0.5 mg per day for those weighing 20 kg or more. After at least four days, increase to 0.5 mg per day for under-20 kg patients or 1.0 mg per day for those 20 kg or heavier, maintaining this dose for no less than 14 days. Doses may then be raised every 2 weeks or more in steps of 0.25 mg per day (under 20 kg) or 0.5 mg per day (20 kg or more). The usual dose range is 0.5 mg to 3 mg per day, and this applies to children and adolescents weighing no less than 15 kg.	FDA-approved for the use shown; label effective 2026-06-25; original approval date not retrieved	FDA label: Efficacy was established in 3 short-term trials in children and adolescents (ages 5 to 17 years). DailyMed label

Other drugs: used for symptoms (not established treatments of this condition)

Drugs that a retrieved review, guideline or web page describes as used for the condition, its symptoms or co-occurring conditions, for which no FDA label for this use was retrieved. Drugs still being tested in trials are listed in the Clinical trials section below.

Drug	Indication	Key evidence
melatonin	address sleep problems in ASD [associated symptom or comorbidity, not the condition itself]	"Melatonin is commonly used to address sleep problems in ASD and appears to be effective." Pharmacological treatments in autism spectrum disorder: a narrative review / Journal of Psychopathology
quetiapine	treatment of irritability and aggressive behavior (population: children and adults with ASD)	"Beside the evidence of efficacy for aripiprazole and risperidone, there are few studies of other atypical antipsychotics such as quetiapine, ziprasidone and paliperidone, although these are still commonly used in clinical practice in the treatment of irritability and aggressive behavior in children and adults with ASD." Pharmacological treatments in autism spectrum disorder: a narrative review / Journal of Psychopathology
ziprasidone	treatment of irritability and aggressive behavior (population: children and adults with ASD)	"Beside the evidence of efficacy for aripiprazole and risperidone, there are few studies of other atypical antipsychotics such as quetiapine, ziprasidone and paliperidone, although these are still commonly used in clinical practice in the treatment of irritability and aggressive behavior in children and adults with ASD." Pharmacological treatments in autism spectrum disorder: a narrative review / Journal of Psychopathology
paliperidone	treatment of irritability and aggressive behavior (population: children and adults with ASD)	"Beside the evidence of efficacy for aripiprazole and risperidone, there are few studies of other atypical antipsychotics such as quetiapine, ziprasidone and paliperidone, although these are still commonly used in clinical practice in the treatment of irritability and aggressive behavior in children and adults with ASD." Pharmacological treatments in autism spectrum disorder: a narrative review / Journal of Psychopathology

Drug	Indication	Key evidence
methylphenidate	reducing hyperactivity and inattention in children with ADHD (population: children with ASD) [associated symptom or comorbidity, not the condition itself]	"Aripiprazole and risperidone have shown effectiveness in treating irritability and aggression in ASD, while methylphenidate may be helpful in reducing hyperactivity and inattention in children with ADHD, although this group may be more susceptible to adverse effects." Pharmacological treatments in autism spectrum disorder: a narrative review / Journal of Psychopathology
Ritalin (methylphenidate hydrochloride)	addresses attention deficits and hyperactivity (population: children with ASD) [associated symptom or comorbidity, not the condition itself]	"Common Medications: Ritalin, Adderall Purpose: Addresses attention deficits and hyperactivity Antidepressant Medications Common Medications: Prozac, Zoloft Purpose: Alleviates symptoms of anxiety and depression Medications such as risperidone were the first FDA-approved drugs for managing irritability associated with autism, particularly for children aged 5 and older." Medications Commonly Used for Autism / Advanced Autism Services
amphetamine-based medications	treat ADHD symptoms in children with autism [associated symptom or comorbidity, not the condition itself]	"Stimulant Medications: Benefits and Drawbacks Doctors commonly prescribe stimulants such as methylphenidate (Ritalin, Concerta) and amphetamine-based medications (Vyvanse, Adderall) to treat ADHD symptoms in children." Best ADHD Medication for Autism: Comprehensive Guide for Parents - Apple ABA
Strattera (atomoxetine hydrochloride)	manage anxiety or mood swings in children with autism (population: children with autism who experience anxiety or mood swings) [associated symptom or comorbidity, not the condition itself]	"For children with autism who experience anxiety or mood swings, non-stimulant medications like Strattera may be a better option." Best ADHD Medication for Autism: Comprehensive Guide for Parents - Apple ABA

Recently approved or updated (FDA, EMA and CDSCO announcements)

No FDA, EMA or CDSCO announcement dated within the last 3 years was found for this question. This does not mean none exists.

Safety warnings (FDA boxed warnings)

Drug(s) (FDA label)	Boxed warning, in the label's words
quetiapine	"INCREASED MORTALITY IN ELDERLY PATIENTS WITH DEMENTIA-RELATED PSYCHOSIS; and SUICIDAL THOUGHTS AND BEHAVIORS Increased Mortality in Elderly Patients with Dementia-Related Psychosis Elderly patients with dementia-related psychosis treated with antipsychotic drugs are at an increased risk of death [see Warnings and Precautions (5.1)]."
ziprasidone	"INCREASED MORTALITY IN ELDERLY PATIENTS WITH DEMENTIA-RELATED PSYCHOSIS Elderly patients with dementia-related psychosis treated with antipsychotic drugs are at an increased risk of death. Ziprasidone is not approved for the treatment of patients with dementia-related psychosis [see Warnings and Precautions (5.1)] ."
paliperidone , Risperidone	"INCREASED MORTALITY IN ELDERLY PATIENTS WITH DEMENTIA-RELATED PSYCHOSIS Elderly patients with dementia-related psychosis treated with antipsychotic drugs are at an increased risk of death. Paliperidone extended-release tablets are not approved for the treatment of patients with dementia-related psychosis [see Warnings and Precautions (5.1)]."
methylphenidate	"ABUSE, MISUSE, AND ADDICTION Methylphenidate hydrochloride extended-release has a high potential for abuse and misuse, which can lead to the development of a substance use disorder, including addiction."
Ritalin (methylphenidate hydrochloride)	"ABUSE, MISUSE, AND ADDICTION Ritalin has a high potential for abuse and misuse, which can lead to the development of a substance use disorder, including addiction. Misuse and abuse of CNS stimulants, including Ritalin, can result in overdose and death [see Overdosage (10)] , and this risk is increased with higher doses or unapproved methods of administration, such as snorting or injection."
Strattera (atomoxetine hydrochloride)	"SUICIDAL THOUGHTS AND BEHAVIORS IN PEDIATRIC PATIENTS 6 YEARS OF AGE AND OLDER All STRATTERA-treated pediatric patients 6 years of age or older should be monitored and observed closely for suicidal thoughts and behavior, clinical worsening, or unusual changes in behavior, especially during the initial months of therapy or at times of dosage changes."
Aripiprazole	"S: INCREASED MORTALITY IN ELDERLY PATIENTS WITH DEMENTIA-RELATED PSYCHOSIS AND SUICIDAL THOUGHTS AND BEHAVIOURS WITH ANTIDEPRESSANT DRUGS Elderly patients with dementia-related psychosis treated with antipsychotic drugs are at an increased risk of death. Aripiprazole is not approved for the treatment of patients with dementia-related psychosis [see Warnings and Precautions (5.1)]."

These are the labels' own boxed warnings, not a complete safety review: read the full label. No boxed warning on the retrieved label of: melatonin. No FDA label could be matched for: amphetamine-based medications.

Clinical trials

Trial	Intervention	Phase	Status (last update)	What the record shows
NCT07435740	Sleep Parent Training (SPT); Sleep Parent Education (SPE)	Phase not applicable	Recruiting (2026-10-08)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT07054866	Speech-Gesture combinations	Phase not applicable	Recruiting (2026-10-08)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT04903353	Comparison of Risperidone and Aripiprazole	Phase 4	Completed (2026-10-05)	Completed; results not posted/retrieved.

Trial	Intervention	Phase	Status (last update)	What the record shows
NCT07257939	NTI164; Placebo	Phase 3	Not yet recruiting (2026-09-30)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT07285798	KarXT; KarX-EC	Phase 3	Not yet recruiting (2026-09-28)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT06229210	Lumateperone	Phase 3	Recruiting (2026-09-25)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT06706674	Lumateperone high dose; Lumateperone low dose	Phase 3	Recruiting (2026-09-25)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT07243561	hUC-MSC-Exos Nasal Spray	Phase 1/2	Recruiting (2026-09-25)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT06690398	Lumateperone high dose; Lumateperone low dose	Phase 3	Recruiting (2026-09-25)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT03279471	Sertraline; CBT/BIACA	Phase 2	Completed (2026-09-23)	Completed; results posted (outcome not retrieved here).

2 further registry record(s) were retrieved and are not shown (judged not to be treatment trials for this question).

Registry status shows that research is being conducted, not that a treatment works. Registry status is separate from published results.

Key studies

1. **Wang et al. (2026)**, meta-analysis: "Exercise significantly improved overall, gross, and fine motor skills in children with ASD." [1]
2. **Hao et al. (2026)**, systematic review: "There was a significant increase in XR autism research after 2020, with virtual reality (VR) as the largest modality (n = 61, 53.0%), followed by augmented reality (AR) (n = 36, 31.3%)." [6]
3. **Restoy et al. (2024)**, meta-analysis: "Studies of non-pharmacological interventions showed significant improvement in both ER and ED." [8]
4. **Sandbank et al. (2023)**, meta-analysis: "When effects were then restricted to exclude those at high risk of detection bias, only one significant summary effect was estimated-naturalistic developmental behavioral interventions on measures of diagnostic characteristics of autism (0.30, 0.03 to 0.57; P=0.03)." [9]
5. **Siafis et al. (2022)**, meta-analysis: "There were some indications of improvement by carnosine, haloperidol, folinic acid, guanfacine, omega-3-fatty-acids, probiotics, sulforaphane, tideglusib and valproate, yet imprecise and not robust." [10]

Conflicting evidence

None was retrieved.

What remains under investigation

- **Efficacy:** Head-to-head trials comparing exercise, diet, and behavioral interventions for specific ASD symptoms (e.g., social communication vs. motor skills).
- **Effectiveness:** Long-term outcomes of caregiver-assisted NDBIs and ketogenic diet interventions in diverse clinical settings.
- **Feasibility:** Scalability of extended reality (XR) interventions for routine clinical use in autism treatment.
- **Association:** Exploration of blood-based biomarkers (e.g., oxidative stress, inflammatory cytokines) as predictors of treatment response in ASD.
- **Exploratory:** Dual-target deep brain stimulation's broader applicability and safety in larger pediatric ASD cohorts.

Evidence limitations

- Findings are drawn from abstracts, registry records and web pages, not full texts; study quality was not appraised from the full papers.
- Doses appear only where an FDA label gives them for this condition (section 2.1); for any other use, see a prescribing source. Approval dates appear only where a retrieved source states them.
- Excluded retracted publication: Efficacy of oral folinic acid supplementation in children with autism spectrum disorder: a (PMID 39243316, 2024). Retracted work is not used as evidence.
- 4 retrieved record(s) were excluded because the relevance reading judged them not to answer the question.

Sources

1. Wang T, Dong X, Jia Y, et al. Dose-response relationship between exercise interventions and motor skills in children with autism spectrum disorder: a systematic review and multilevel meta-analysis. *European journal of pediatrics*. 2026-09-18. [PMID 42758334](#). DOI 10.1007/s00431-026-07421-4.
2. Liu L, Zhao Q, Zhou J, et al. Evaluating the therapeutic efficacy of a modified ketogenic diet in children with autism spectrum disorder: a randomized controlled trial. *BMC pediatrics*. 2026-09-11. [PMID 42763398](#). DOI 10.1186/s12887-026-07708-3.
3. Seng GJ, Lin JY, Huang WL, et al. Feasibility and Preliminary Efficacy of a Caregiver-Assisted Group-Based Naturalistic Developmental Behavioral Intervention Program for Autistic Preschoolers: A Pilot Randomized Controlled Trial. *Autism : the international journal of research and practice*. 2026-07-23. [PMID 42494078](#). DOI 10.1177/13623613261462574.
4. Jacob US, Maguvhe MO, Pillay J, et al. Psychological interventions and mental health outcomes among caregivers of children with intellectual disability: a systematic review and meta-analysis. *Frontiers in psychology*. 2026-08-18. [PMID 42682640](#). DOI 10.3389/fpsyg.2026.1883919.
5. Fisberg M, Nogueira-de-Almeida CA, Konstantyner T, et al. FEEDSETUP: feeding difficulties spectrum evaluation and treatment up-to-date proposal. *Jornal de pediatria*. 2026-09-18. [PMID 42716484](#). DOI 10.1016/j.jped.2026.101610.
6. Hao D, Mansi H, Elkefi S, et al. Extended reality (XR) interventions in supporting children with autism: A systematic review. *International journal of medical informatics*. 2026 Dec. [PMID 42784985](#). DOI 10.1016/j.ijmedinf.2026.106719.
7. Shi M, Hu J, Pei J, et al. Dual-target deep brain stimulation for severe pediatric autism: Sustained core symptom reduction in a 12-month case study. *Clinical neurophysiology : official journal of the International Federation of Clinical Neurophysiology*. 2026 Dec. [PMID 42759099](#). DOI 10.1016/j.clinph.2026.2112403.
8. Restoy D, Oriol-Escudé M, Alonzo-Castillo T, et al. Emotion regulation and emotion dysregulation in children and adolescents with Autism Spectrum Disorder: A meta-analysis of evaluation and intervention studies. *Clinical*

psychology review. 2024 Apr. [PMID 38401510](#). DOI 10.1016/j.cpr.2024.102410.

9. Sandbank M, Bottema-Beutel K, Crowley LaPoint S, et al. Autism intervention meta-analysis of early childhood studies (Project AIM): updated systematic review and secondary analysis. *BMJ (Clinical research ed.)*. 2023 Nov 14. [PMID 37963634](#). DOI 10.1136/bmj-2023-076733.
10. Siafis S, Çıray O, Wu H, et al. Pharmacological and dietary-supplement treatments for autism spectrum disorder: a systematic review and network meta-analysis. *Molecular autism*. 2022 Mar 4. [PMID 35246237](#). DOI 10.1186/s13229-022-00488-4.

Other sources (regulators, guideline organisations, web pages, FDA labels)

- [Pharmacological treatments in autism spectrum disorder: a narrative review | Journal of Psychopathology](#)
- [Medications Commonly Used for Autism | Advanced Autism Services](#)
- [Best ADHD Medication for Autism: Comprehensive Guide for Parents - Apple ABA](#)
- [FDA drug label: Aripiprazole \(label effective 2026-03-05\)](#)
- [FDA drug label: Risperidone \(label effective 2026-06-25\)](#)

Research support, not a substitute for clinical judgement. All citations, trial IDs, links and numbers in this report were checked against the retrieved sources.